IMPACT OSAKA HACKATHON 2026 — SUBMISSION DOSSIER

OSAKA VOICEBRIDGE AI

Multilingual Voice-First Accessibility City Assistant for Osaka, Japan

Tagline: *"One city. Every language. No one left behind."*

Category: Social Impact & Accessible Mobility	**Target Location:** Osaka, Kansai Region, Japan
Repository: github.com/bhagwatdhonde15/OSAKA-VOICEBRIDGE-AI	**Tech Stack:** React 19, FastAPI, ElevenLabs AI
Author / Lead: Bhagwat Dhonde	**Date:** October 2026

1. Executive Summary

OSAKA VOICEBRIDGE AI *is an intelligent, voice-first multilingual mobility assistant created specifically for the* ***Impact Osaka Hackathon 2026****. It addresses the acute accessibility and language barriers encountered by over* ***28 million international tourists*** *and* ***growing elderly populations*** *navigating Osaka's sprawling transit network. By fusing instant speech recognition, real-time ElevenLabs neural speech generation, a deterministic transit reasoning engine, and oversized Japanese staff flashcards, VoiceBridge AI ensures every traveler can independently find step-free elevators, understand transit directions, and dine safely across Osaka.*

2. The Osaka Challenge: The Problem We Solve

Osaka is Japan's vibrant economic and cultural hub, yet its transit infrastructure presents severe obstacles for visitors:

- **The 'Umeda Dungeon' & Transit Labyrinth:** Umeda/Osaka Station is one of the world's most intricate underground networks with 6 interconnected rail operators and over 1,200 shops, causing high disorientation for travelers with luggage or wheelchairs.
- **Severe Language Barrier:** Essential platform elevator signage and station staff announcements are predominantly in Japanese Kanji/Kana, leaving non-Japanese speakers stranded during transit transfers.
- **Elderly & Reduced Mobility Access:** Japan's aging demographic and elderly tourists require stair-free paths, resting zones, and slower, clearer spoken instructions.
- **Food Allergen Safety:** Osaka's famous street foods (Takoyaki, Okonomiyaki) frequently utilize dashi (fish broth) and shrimp powder. International tourists with severe allergies face life-threatening communication gaps at busy outdoor stalls.

3. The 3 Official Hackathon Demo Scenarios

OSAKA VOICEBRIDGE AI features 3 fully verifiable, 1-click demonstration scenarios designed for judge evaluation:

Scenario	User Spoken Query	AI Intent & Action	Verifiable Output
1. Finding Osaka Station *(Step-Free Route)*	“I am looking for Osaka Station. Can you guide me through the elevator route with no stairs?”	**Intent:** station_navigation_guidance **Confidence:** 99% **Action:** Step-free elevator routing	Guides user to Car 6 on Midosuji Line $\rightarrow$ Umeda M16 Central Elevator $\rightarrow$ Central South Gate with 0 stairs.
2. Simpler Directions *(Cognitive Simplicity)*	“I feel overwhelmed by the transit maps. Please give me simpler travel directions for the Midosuji subway and ICOCA card.”	**Intent:** simplified_transit_instructions **Confidence:** 97% **Action:** 4-step progressive disclosure	Bite-sized directions: 1) Find Red 'M' line, 2) Buy/tap ICOCA at pink machine, 3) Board Platform 1, 4) Exit at M16.
3. Essential Phrase *(Allergy & Staff Card)*	“How do I ask if Takoyaki has shrimp or shellfish? I need to show a clear Japanese card to the street food vendor.”	**Intent:** dietary_allergen_verification **Confidence:** 98% **Action:** Bilingual vendor flashcard	Synthesizes Japanese phrase + pops up high-contrast Kanji flashcard for food stalls: ■■■■■■■■■■■■■■■■■■■■■■■■■■...

4. AI Decision Matrix & Observability Panel

A critical hackathon requirement is demonstrating explainable, transparent AI reasoning without hallucinations. The interface incorporates a dedicated **AI Decision Matrix** that exposes real-time engine telemetry:

Telemetry Metric	Value / Implementation	Safety & Certainty Guarantee
Intent Classification	Classified into 5 distinct categories: navigation, transit, dietary_allergy, emergency, phrases.	Strict deterministic routing prevents hallucinated transit paths.
Confidence Score	97% to 99% evaluated confidence based on multilingual regex token matching.	High confidence required before dispatching synthesized voice audio.
Multilingual Detection	Real-time classification of English, Japanese (Kanji/Kana), Hindi (Devanagari), and Spanish.	Respects traveler's preferred language override.
Uncertainty Advisory	Transparently surfaces live transit status disclaimers and recommended platform confirmations.	Honest indicator flag: static verified dataset vs. simulated live feed.

5. Technical Architecture & System Design

OSAKA VOICEBRIDGE AI is built on a clean decoupled architecture prioritizing rapid execution, rock-solid stability, and zero external dependency failure during judging:

Layer	Technology	Role & Key Implementation
Frontend UI	React 19, TypeScript, Vite 8, Tailwind CSS	Responsive Japanese dark luxury aesthetic (#080B16), reactive VoiceOrb, Leaflet OpenStreetMap interactive station map, and audio controls.
Voice AI Engine	ElevenLabs Neural API + Web Speech API	Server-side ElevenLabs synthesis (eleven_multilingual_v2) with transparent, reliable client-side Web Speech API fallback in demo mode.
Backend API	Python 3.12 / 3.14, FastAPI, Uvicorn	High-performance asynchronous endpoints (/api/assistant/query, /api/voice/synthesize, /api/stations, /api/scenarios).
Decision Core	DecisionEngine (decision_engine.py)	Multilingual classification, accessibility constraint analysis (wheelchair, senior), and step-free transit route resolution.
Local Supervisor	serve.py & start_voicebridge.bat	Clean process manager: clears stale ports 8000 & 5173, starts both servers, monitors health, and auto-opens default browser.

6. ElevenLabs Voice AI Integration

Voice is the fastest, most humane interface for travelers carrying luggage or navigating bustling transit hubs. Our integration implements full compliance with hackathon voice requirements:

- **Server-Side Security:** Credentials (ELEVENLABS_API_KEY, ELEVENLABS_VOICE_ID) reside exclusively in backend environment variables, never exposed in client code or Git.
- **Streaming Synthesis:** Post request to /api/voice/synthesize passes synthesized text to ElevenLabs multilingual model, streaming base64 audio back for instant playback.
- **Honest Offline Fallback:** When API credentials or credits are depleted, the platform seamlessly switches to browser-native Web Speech API, clearly labeled as Demo Mode without falsifying ElevenLabs branding.
- **Senior Speed Control:** Spoken pace can be dialed to 0.8x with high-clarity enunciation for older visitors.

7. Comprehensive Accessibility Features

In alignment with Japan's Universal Design standards and the Osaka 2025/2026 inclusive tourism guidelines:

Feature	Benefit to Visitor
Wheelchair & Step-Free Mode	Filters routes exclusively through verified elevators, level boarding ramps, and wide automated gates (e.g., Umeda Gate 3).
Senior Mode (A+)	Expands typography by +30%, highlights bench rest zones (Toki-no-Hiroba Plaza 5F), and reduces voice response speed.
High-Contrast Mode	Provides WCAG AAA contrast ratio compliance, eliminating visual glare for low-vision users.
Station Staff Flashcard	Generates an oversized modal with Japanese text and phonetic Furigana so travelers can simply show their screen to train conductors or vendors.

8. Scaling Roadmap & National Vision

Osaka VoiceBridge AI is architected to scale beyond Osaka into a ubiquitous national accessibility standard across Japan:

Phase	Timeline	Scope & Milestones
Phase 1: Osaka Metro Core	Q4 2026	Deploy across all 133 Osaka Metro stations, Kansai International Airport (KIX) express lines, and Dotonbori food district.
Phase 2: Kansai Golden Route	2027	Expand to Kyoto Municipal Subway, Hankyu/Hanshin intercity rail, and JR West Tokaido Shinkansen bullet train hubs.
Phase 3: National Smart Tourism	2028+	Integration with Japan National Tourism Organization (JNTO), digital Suica/ICOCA transit wallets, and international mega-event venues.

9. Hackathon Judging Rubric & Evaluation Alignment

OSAKA VOICEBRIDGE AI is engineered to maximize scoring across the official Impact Osaka Hackathon evaluation dimensions:

Judging Dimension	Evaluation Evidence & Project Strengths	Readiness
Social Impact & Inclusivity	Directly tackles the 28M+ tourist language barrier and aging society mobility in Osaka. Provides step-free navigation, cognitive transit simplification, and life-saving Takoyaki food allergen safety cards.	**High Impact**
Voice AI & ElevenLabs	Mandatory ElevenLabs text-to-speech integration implemented server-side with eleven_multilingual_v2 model. Includes transparent, honest client-side Web Speech API fallback for zero-downtime judging.	**Compliant**
Technical Architecture	Production-ready decoupled stack: React 19, TypeScript, Vite 8, Python FastAPI. Includes explainable AI Decision Matrix with 97%-99% confidence scores and zero hallucinated transit directions.	**Production**
Design Craft & Polish	Venture-backed dark luxury aesthetic (#080B16, #11162A, #8B5CF6, #67E8F9). Features interactive Leaflet Osaka Metro map, reactive VoiceOrb, and Senior A+ font scaling.	**Polished**

10. Hackathon Submission & Deliverables Index

Resource	Link / Location	Status
GitHub Repository	https://github.com/bhagwatdhonde15/OSAKA-VOICEBRIDGE-AI	**Public & Synced**
Official Competition Video	output/OSAKA_VOICEBRIDGE_AI_COMPETITION_VIDEO.mp4 (90s HD)	**Rendered (20.2 MB)**
Local Web Application	http://localhost:5173 (React Vite Frontend)	**Active (HTTP 200)**
FastAPI Health Check	http://127.0.0.1:8000/api/health	**Active (Healthy)**
API Interactive Documentation	http://127.0.0.1:8000/docs (Swagger UI)	**Active**

OSAKA VOICEBRIDGE AI — Impact Osaka Hackathon 2026

“One city. Every language. No one left behind.”

Lead Developer: Bhagwat Dhonde | Fully Verified & Ready for Evaluation